# DESKLESS

## Inspection reports, without the desk.

Walk the place, photograph each issue, say what is wrong. The phone writes the report, with no network.




Team techazsure | iQOO Hackathon 2026 Grand Finale | Productivity track | Bengaluru, 9 to 11 October 2026

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# Every place is walked and written up

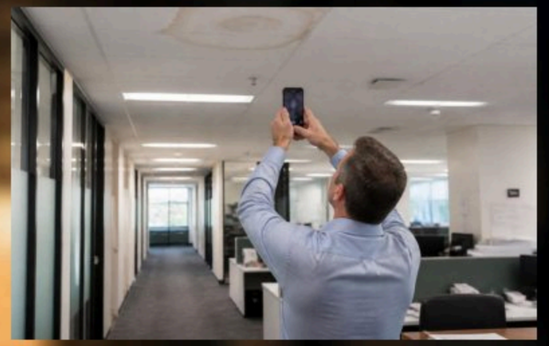
Someone walks every kind of place every day and has to write down what is wrong with it. Construction is one of them, not the centre.



Hospitals and clinics



Colleges and campuses



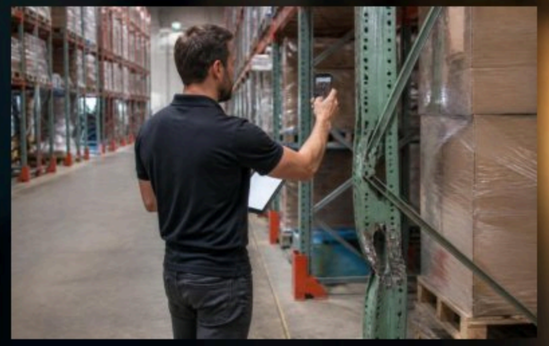
Government and company offices



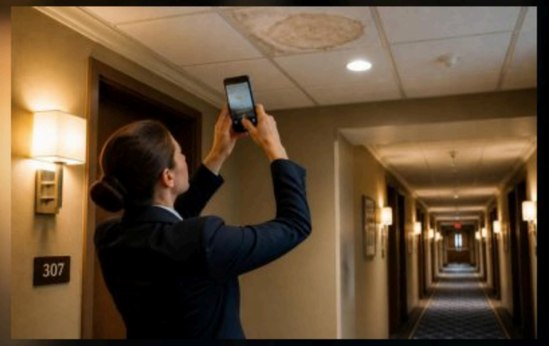
Apartment blocks and housing societies



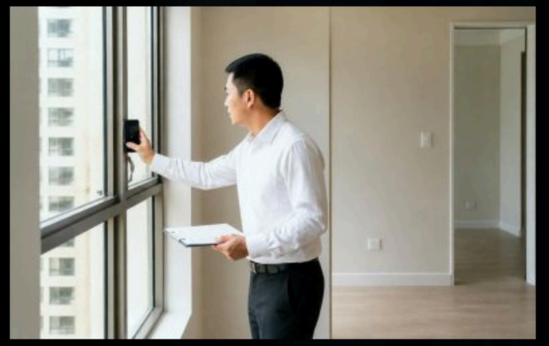
Malls and retail floors



Factories and warehouses




Hotels and hostels



Construction and handovers, one of many

03

# The problem: the walk is one hour, the desk is another

What goes wrong today between seeing an issue and fixing it.




On the walk: issues are found in wards, corridors, basements and plant rooms, often with no network.

## ANOTHER HOUR

The write-up happens at a desk hours later, from memory and loose photos.

Photos sit in WhatsApp groups with no location, owner or deadline attached.

Forms want typed English on a small screen; people think and speak in Hindi, Kannada, Tamil or a mix.

Nobody checks whether the photo really shows what the note says.

The person who has to fix it hears last; management sees nothing until the report is typed.

04

# Why it is needed, and how much

Scale of the problem in India, with the sources named.

## USD 15.5 BN

India's integrated facility management market in 2025, heading to USD 24.9 bn by 2031 at about 8.3% a year (Mordor Intelligence)

## ~69,000

Hospitals in India (Statista estimate), where rounds, medication audits and expiry checks are daily routine

## 47,800+

Universities and colleges in India: 1,213 and 46,624 (AISHE 2022-23), each inspected for safety, hygiene and infrastructure

## EVERY DAY

Someone in each of these places walks, looks and writes; that report is the only record. The cost is paid in hours at the desk, issues never written down, and repairs that start late

05

# Existing solutions and their flaws

What places use today, and what each of them misses.

| # | EXISTING SOLUTIONS | WHAT EACH MISSES |
|---|---|---|
| 01 | Paper registers and WhatsApp groups (free and familiar) | Nothing is structured, tracked or searchable; the photo has no location, owner or deadline |
| 02 | Cloud inspection apps (SafetyCulture, UpKeep, Fieldwire and alike) | Tap-through forms in typed English; about USD 24 per seat per month; offline mode only edits forms already on the device, and reports and notifications wait for the network |
| 03 | Cloud AI add-ons: voice notes and photos processed on a server | The voice note and the photo leave the premises, which hospitals, government offices and housing societies cannot accept; and nothing works where there is no network |
| 04 | All of them, for the people who use the place | None lets students, residents, patients and visitors report an issue in their own words and hear back when it is fixed |

06

# Our solution: say it, the phone writes it

Photo plus voice in. Finished report out. All on the phone.

**PHOTO + VOICE IN**

**SIGNED PDF OUT**

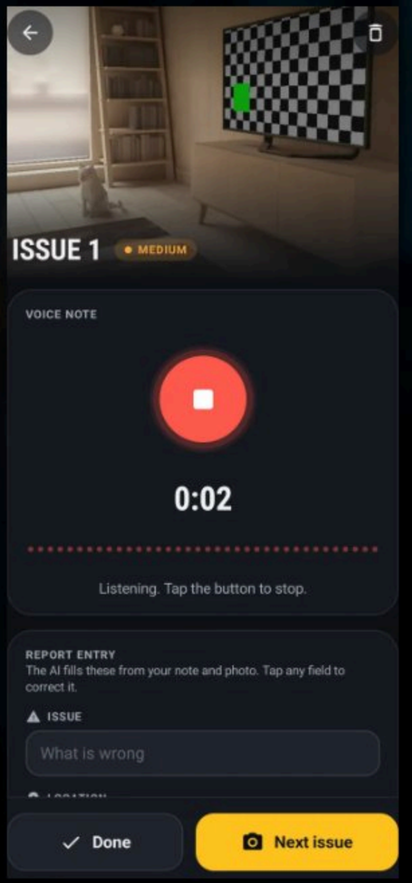
Photograph the issue or film a 20-second clip, and say what is wrong in Hindi, English or a mix.

Whisper writes English text on the phone; Gemma 4 fills issue, location, severity, owner, due date and type, and says whether the photo matches the note.

A branded English PDF with the issue map, sensor readings, per-owner action items and a fingerprint signature, built in seconds.

Works in airplane mode end to end: 3 to 11 s from the end of the recording to filled fields on the iQOO 15.

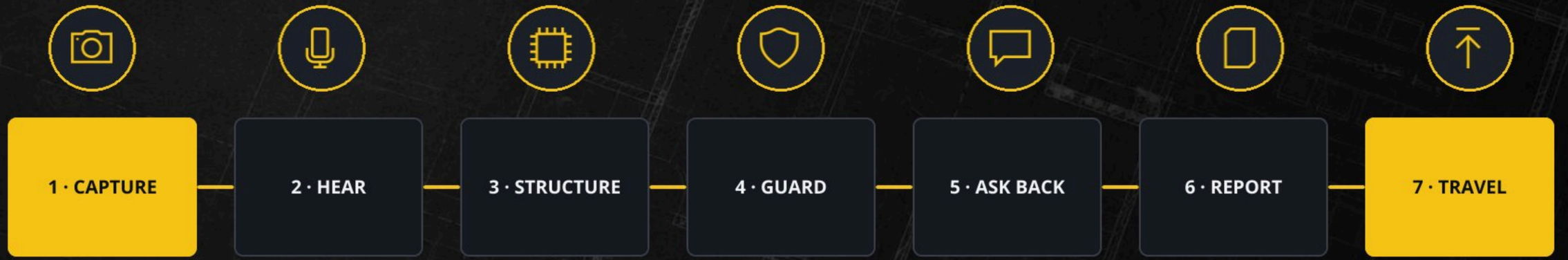
With a network, the report reaches the organisation's server and every role's console.



07

# How it works: from a voice note to a repair

Seven steps. The first five happen entirely on the phone; nothing is uploaded to make a report.




**1 · CAPTURE**

Photo or 20-second clip plus a voice note. Position, compass, light and step count stamped by the phone.

**2 · HEAR**

Whisper small on the phone turns Hindi, English or mixed speech into English text. Long notes split at pauses.

**3 · STRUCTURE**

Gemma 4 E2B on the phone's GPU reads the note and the photo, fills the five fields and judges whether the photo matches.

**4 · GUARD**

Plain code checks every value against what was said or seen. Anything else is held and shown, never written silently.

**5 · ASK BACK**

If where, who or by when is missing, the phone asks one question aloud and listens for the answer.

**6 · REPORT**

A signed PDF plus CSV, calendar and per-owner task list, and an AI review of the whole report.

**7 · TRAVEL**

To the server over HTTPS, phone to phone sealed when there is no network, or to a laptop through iQOO Office Kit.

08

# Inside the phone: the AI that stays on the device

Technical depth on the on-device models and the rules around them.

ON THE PHONE, NOTHING UPLOADED

Whisper small via sherpa-onnx with Silero VAD: Hindi + English by default; also Kannada, Tamil, Telugu, Marathi and Bengali.

Gemma 4 E2B (2.6 GB) via LiteRT-LM on the Adreno GPU through OpenCL, CPU fallback: photo and text in, strict JSON out.

Measured on the iQOO 15: 3 to 11 s from the end of the recording to filled fields.

THE RULE
The model words things and fills forms; plain code decides and checks. Every field carries its source: heard, seen, phone, known, asked, told, judged or typed.

GUARD AND MEMORY
A location, owner or deadline is written only if its words were said or seen; otherwise it is held and shown. A phone-side memory learns owners and places; it is never put in the prompt.

ASK, TELL, MEASURE

Ask or tell: spoken instructions (mark issue 3 high, give it to Suresh, due Friday) are planned by code and applied only on Do it, with Put it back.

Scorecard: 20 fixed notes, spoken aloud by the phone's own voice and heard by Whisper; on the five-note run 24 of 25 fields right and 0 made up.

Models fetched by the app over Wi-Fi (2.96 GB, checksummed) or imported from a folder on a phone with no internet.

09

# The phone as an instrument

Sensors used as evidence, stamped without taps.

STAMPED WITHOUT TAPS
Compass direction, camera aimed overhead or at the ground, light in lux, steps since the last issue, satellite position on an issue map drawn with no map tiles.

INSTRUMENTS, ONE TAP EACH
Level and plumb from the accelerometer, noise in dB from the microphone, vibration RMS and dominant frequency by FFT.

NFC tags name the place once; lift-to-talk through the proximity sensor; the torch runs off the light sensor. Fingerprint sign-off: an ECDSA P-256 key in the Android Keystore signs the finished PDF, and the signature is re-checked wherever the report lands. Readings go to the model with the note and into the PDF as a guide, marked as not calibrated.



# When there is no network

Offline-first paths for the report.



AIRPLANE MODE
Capture, transcription, structuring and the PDF all happen on the phone. Nothing waits for a signal: a ward, a basement or a plant room is no different from the office.

PHONE TO PHONE
Nearby Connections over Bluetooth and Wi-Fi Direct. Every copy is sealed for one phone (ECDH P-256, HKDF, AES-256-GCM); relay phones carry what they cannot read; a signed receipt travels back.

WALKIE-TALKIE AND HAND-OVER
Hold to talk: the clip rides the same sealed path. A user's entry with no network is handed to a nearby staff phone and posted when that phone finds a network.

OFFICE KIT AND UPLOAD
The PDF goes to the laptop by file transfer and the action items by the shared clipboard. Everything queued goes by itself the moment a network returns.

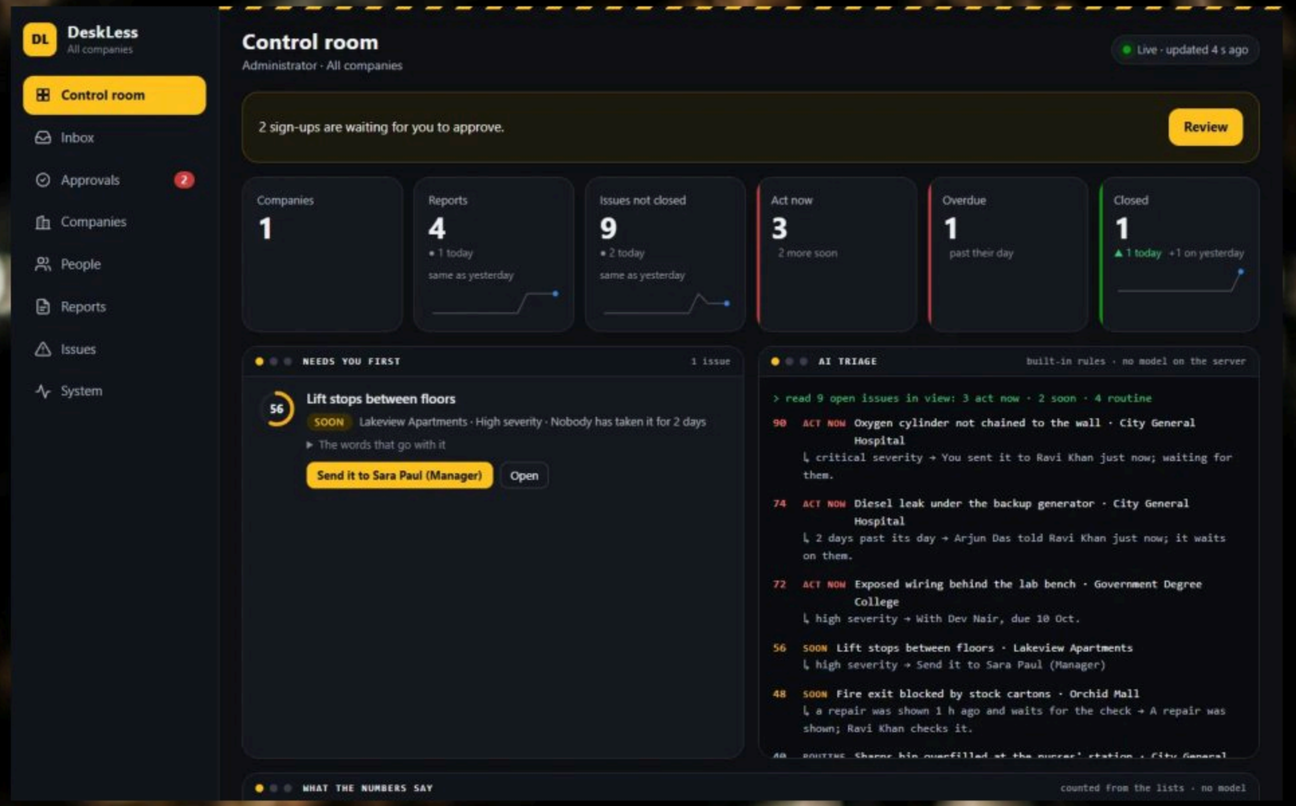
# From report to repair: the server and the consoles

What happens once a report reaches the organisation.

ROLES AND WORKFLOW
Five roles: administrator, managing director, manager, inspector, technical. Sign-up with an emailed code, approval by the level above, a unique DeskLess ID. Open → assigned with a due date → fixed with a repair photo compared by the model → closed, or sent back.

THE ASSISTANT
Ranks every issue 0 to 100 (Act now, Soon, Routine) from severity, deadline, time untaken, danger words, photo mismatch and repeats at one place; names up to three people who could take it, with reasons; proposes the next step as a button. Its own rounds chase untaken and late work by email.

CONSOLES AND THE LIVE LINKAGE
A console per role, an inbox, what the numbers say in plain sentences, CSV export, live refresh. Every phone, person, place and service drawn as nodes in 3D, with each report, hand-off and answer flying between them as it happens.



# The people who use the place close the loop

The DeskLess User edition and My desk.

DESKLESS USER
Students, residents, patients and visitors pick the kind of place, then the place, and say what is wrong. The phone writes the entry, reads it back and listens: send it, make it high, add a photo, it is a question. The place's inspector gets it on their desk; the answer comes back to the person's phone and is read aloud.

MY DESK
A staff member's work, said aloud and answered by voice: give the first one to Arjun by Monday; I fixed it (the camera opens and the repair photo is compared); close job 5. Measured through the live server: an entry on the manager's desk in about 10 s, an instruction understood on the phone in 1.5 to 1.8 s.



# Architecture

Three parts. The phone makes the report with no help; the server stores, ranks and routes; the web shows the work and lets each role act on it.

## THE PHONE · makes the report

- **Capture** — photo, clip, voice, sensors
- **Whisper small** — speech to English, 6 languages
- **Gemma 4 E2B on GPU** — fills fields, checks the photo
- **Guard + memory** — code decides, model words
- **Sensors + NFC** — level, noise, light, place tags
- **Signed PDF pack** — Keystore ECDSA, CSV, ICS
- **Sealed relay** — Nearby, AES-256-GCM, receipts
- **Outbox** — uploads when online

**no network needed for any of this**

Kotlin · Jetpack Compose · sherpa-onnx · LiteRT-LM · CameraX · Nearby Connections

**iQOO 15: 3 to 11 s from the end of the voice note to filled fields**

Two editions from one code base: DeskLess (staff) and DeskLess User (the public)

**Phone to phone with no network**
Bluetooth, Wi-Fi Direct; sealed for one phone, relayed, receipted

iQOO Office Kit: PDF to the laptop by file transfer, action items by clipboard

HTTPS → upload

long poll ← to desk

## THE SERVER · stores, ranks, routes

- **nginx** — under a path of the existing site, HTTPS
- **FastAPI + SQLAlchemy** — accounts, roles, reports, issues, workflow
- **PostgreSQL**
- **Assistant** — rank, route, next step
- **Writer, optional** — Gemma 3 1B, llama.cpp
- **Autopilot rounds** — chases late work
- **Mail** — SMTP, codes, hand-offs
- **Metrics + pulses** — requests, health, every hand-off for the live linkage

Python 3.12 · PostgreSQL · Docker Compose · deploy/setup.sh

**Email to people**
sign-up codes, issues passed on, the assistant's rounds

110 Android unit tests · 99 server tests · on every push

JSON → long poll

actions ← assign, close

## THE WEB · shows and acts

- **Home page + sign-up** — film, deck, screens, the way in
- **A console per role** — MD, manager, inspector, technical
- **Inbox · issues · reports** — search, CSV export, live refresh
- **Report types + public door** — a company's own types
- **Live linkage** — every node and hand-off in 3D

Plain HTML, CSS and JS · no build step

**One command**
docker compose up builds both APKs and runs it all

GitHub Actions builds and tests the same tree

# Tech stack

| LAYER | CHOICE |
| --- | --- |
| **App** | Kotlin 2.2, Jetpack Compose, one APK; two editions (staff and user) from one code base; minSdk 24 |
| **Speech and voice** | Whisper small via sherpa-onnx 1.13, Silero VAD, Android text-to-speech; Hindi, English, Kannada, Tamil, Telugu, Marathi, Bengali |
| **Language and vision model** | Gemma 4 E2B (.litertlm) via LiteRT-LM 0.18 on the OpenCL GPU with CPU fallback; Gemma 3 1B via llama.cpp on the server (optional) |
| **Sensors, security, phone to phone** | SensorManager, NFC, BiometricPrompt, Keystore ECDSA P-256, AES-256-GCM sealing; Nearby Connections (P2P_STAR), ZXing QR pairing, HMAC proof |
| **Server, web and ops** | Python 3.12, FastAPI, SQLAlchemy, PostgreSQL, scrypt, HS256 tokens; plain JS and Canvas 2D; Docker Compose, Caddy, nginx, GitHub Actions; 110 Android and 99 server tests |

# What makes DeskLess different

Uniqueness and innovation.

01

### AI on the phone, in Indian languages

Nothing is uploaded to make a report. Hindi, English and four more languages, heard and written on the device.

02

### Evidence over eloquence

The model words, plain code decides. Every field says where it came from; what was not said is held, not invented.

03

### A sealed relay between phones

Reports and voice messages cross phones that cannot read them, with signed receipts back to the sender.

04

### The phone as an instrument

Compass, level, noise, vibration, light, NFC tags and a fingerprint signature, stamped as evidence.

05

### Both sides of the place

The inspector's rounds and the public's voice meet on one desk that talks back, in a hospital, a college or a housing society alike.

06

### An assistant that explains itself

Ranks, routes and chases with its reasons on screen, and a 3D live view of the whole system as it runs.

# Feasibility: it runs today

Proof that it is built, with measured numbers.

**3–11 s**

**110 + 99**

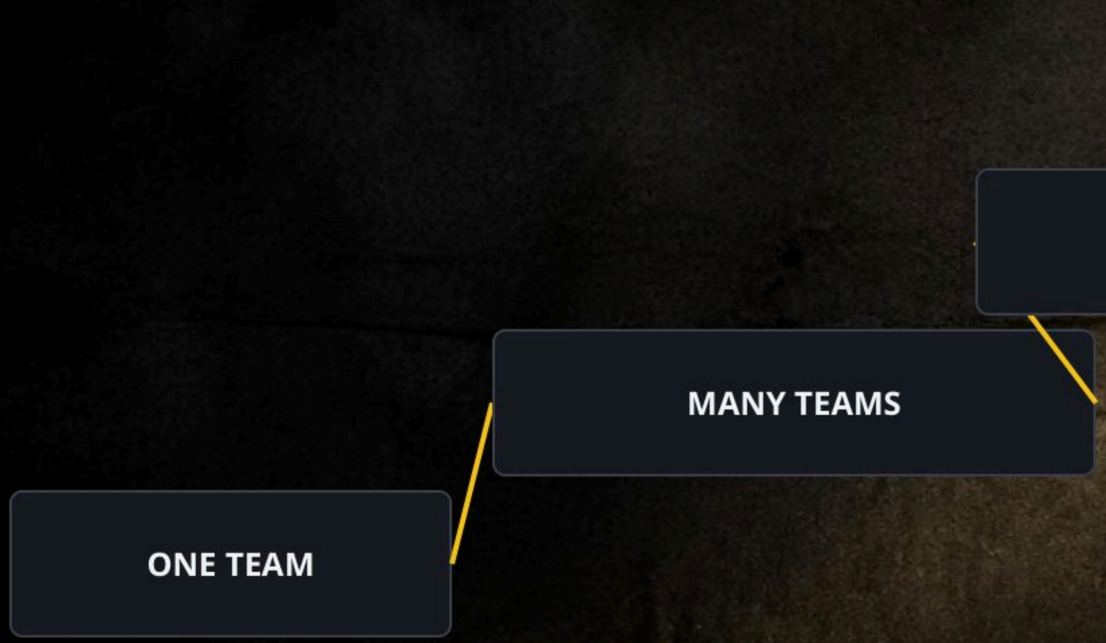
SECONDS FROM VOICE NOTE TO FILLED FIELDS ON THE iQOO 15 · ANDROID AND SERVER TESTS PASSING ON EVERY PUSH

Built in the 48-hour finale as a native Android app plus server; the repository is public on GitHub. Runs on the iQOO 15 with Gemma on the Adreno GPU, and on a Pixel 9a. Live at shaikalthaf.in/competitions/iqoo with sign-up, consoles, assistant and live linkage, deployed with one script beside an existing website. One command, docker compose up, builds both APKs, runs every test and serves it all.



# Scalability

How it grows from one team to many organisations.




- Any kind of place: twelve report types built in (general, hospital, campus, office, apartment, mall, factory, warehouse, hotel, medication audit, community, construction), plus a company's own, pushed to phones at check-in
- Multi-tenant by design: each organisation has its own roles, people, report types and public door
- AI inference scales with phones already in people's hands; the server only stores, ranks and routes
- Models fetched once per phone (2.96 GB); no per-report cloud AI cost
- Stateless FastAPI behind nginx with PostgreSQL: add instances behind the same path; long polls hold no database connection
- Six Indian languages today; more by swapping the speech model file

# Practicality and adoption

Why an organisation can start on Monday.



START ON MONDAY
Nothing to buy: a mid-range Android with 8 GB runs it; the demo runs on a Pixel 9a.
Nothing to learn: walk, photograph, speak; the phone asks when something is missing.
Starts small: one managing director signs up, invites a manager with an eight-character code, and inspectors and technicians join under them.

FITS HOW PLACES ALREADY WORK
The PDF, CSV, calendar file and per-owner task list go wherever reports go today, including by Office Kit to a laptop. Privacy by construction: voice and photo are processed on the phone, identity numbers are encrypted and shown masked, the report is signed. It keeps working where the network does not: wards, basements, plant rooms, new sites.

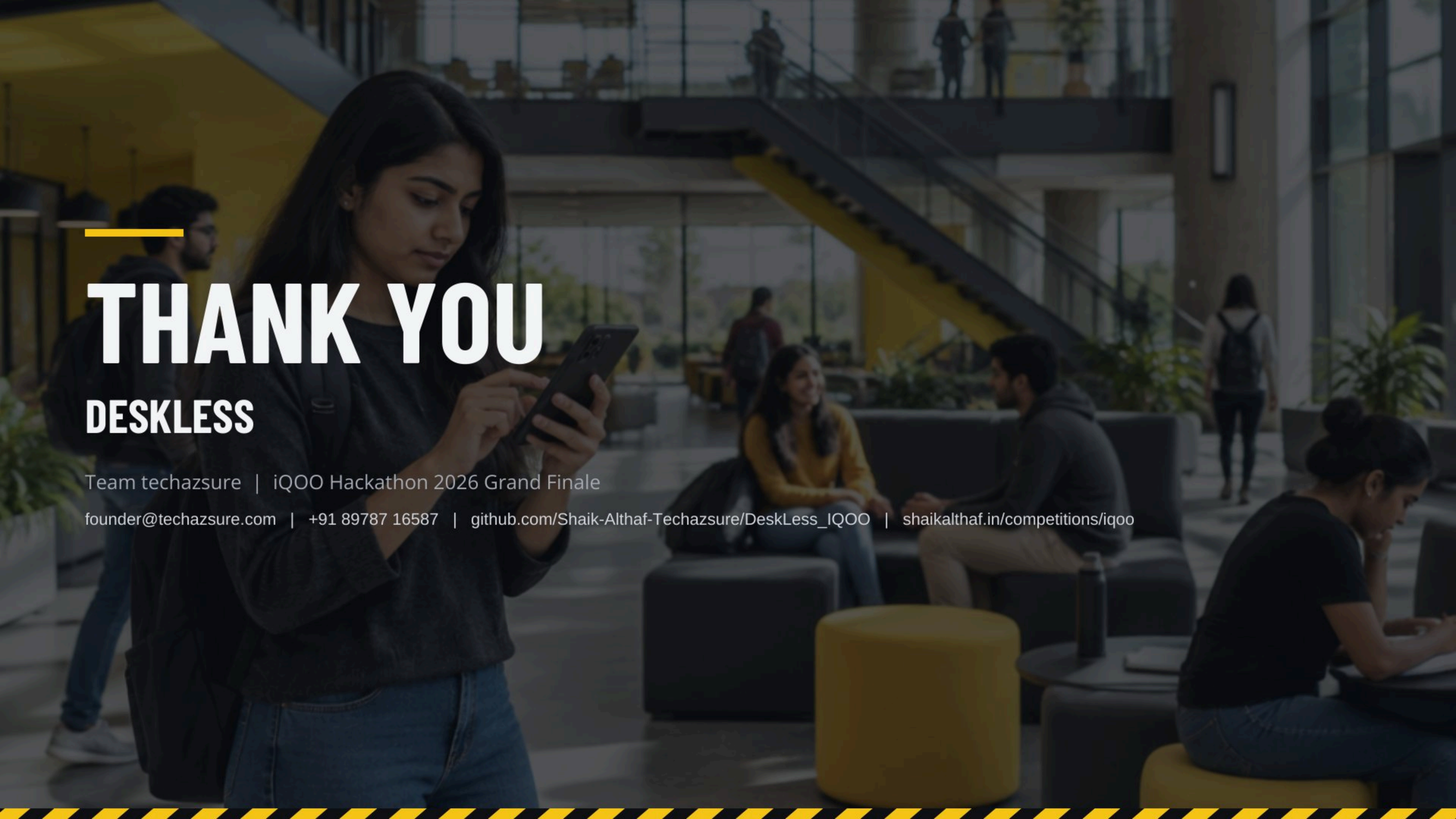
# Where it goes next

Roadmap, three horizons.

NEXT 3 MONTHS
Pilots in one hospital, one college and one housing society. Multi-hop relay and background relaying. A push service so desks announce while the app is closed.

NEXT 12 MONTHS
iOS edition. More Indian languages. Office Kit integration when an SDK is published. Analytics across organisations, with consent.

BUSINESS MODEL
Free for small teams; a per-place subscription for consoles, assistant, server and report types; a self-hosted enterprise licence with support. The on-phone AI stays free: it costs the organisation nothing to run. Full Business Model Canvas in the companion document.



# THANK YOU

## DESKLESS

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